

Li-Syue (Michael) Hsiao

DIGITAL FABRICATION · ROBOTIC TIMBER MACHINING · COMPUTATIONAL DESIGN

v8i69g@gmail.com

Summary

Digital fabrication engineer and co-founder working at the intersection of mass timber construction and robotic automation. Built and operate a KUKA robotic machining line that integrates parametric design (Rhino/Grasshopper) directly with production — from automated BOM generation to KRL tool-paths. Track record spans regulatory certification (Taiwan MOI fire-resistance approvals for CLT), international supply-chain development (Stora Enso, HASSLACHER, Rotho Blaas), and delivery for clients including Siemens Gamesa. Seeking digital fabrication, robotics application, or computational design (VDC) roles in North America.

Work Experience

CCI Co., Ltd.

Chiayi, Taiwan

CO-FOUNDER & VICE PRESIDENT

Jul. 2023 – Present

- Co-founded a mass timber digital fabrication and engineering firm within the Fure Jang group, created to localize robotic timber machining capability in Taiwan; lead a 6-person team spanning fabrication, engineering, and international procurement.
- Designed and commissioned a KUKA robotic machining cell for heavy timber components (Apr. 2024); cut per-component machining time from 3 hours to 30 minutes (6x) and reduced machining error rate from 8% to 1% on the Chiayi Railway Elevation timber structures.
- Key fabrication vendor to Siemens Gamesa Taiwan since 2024: deliver ~50 car-sized horizontal pre-assembly units per year supporting offshore wind turbine tower installation.
- Led certification program earning Taiwan Ministry of the Interior new-construction-material approvals for CLT panels (2025), including 1-hour fire-resistance and wind-driven-rain performance ratings.
- Secured Taiwan distributorships from Stora Enso and K&S; established long-term partnerships with HASSLACHER, XLAM Dolomiti, Rotho Blaas, SIGA (Switzerland), and BG&E (Australia) spanning Europe, Australia, the Americas, and Japan.
- Represent the Fure Jang group in international supplier negotiations covering raw materials, glulam/CLT, structural hardware, waterproofing systems, and protective coatings.
- Built a Claude API × Rhino 8 integration in Python (RhinoCommon): natural-language prompts generate scripts that are previewed, validated, and executed on live models — bringing AI automation into daily modeling and documentation workflows.

TAKEN Construction Material Solutions

Taipei, Taiwan

DIRECTOR OF DIGITAL FABRICATION

Sep. 2021 – Jan. 2023

- Directed digital-geometry workflows for metal building envelopes, integrating façade, fire-protection, and fabrication data into 3D models for doubly-curved surfaces — concurrently with full-time graduate study.
- Project-managed multiple envelope packages with on-site construction supervision, including the Swire Motors service center (2021) and the National Taiwan University Health Center skybridge (2022).

Fure Jang Enterprise Co., Ltd.

Taiwan

ASSISTANT ENGINEER → PROJECT ENGINEER

2018 – 2021

- Drove company-wide 3D/BIM transformation: replaced expert-dependent manual quantity takeoffs (1–2 hours per BOM, non-standardized) with Grasshopper-automated BOM generation at modeling time, cutting takeoff errors from 10% to 2%; the system remains in production use today (2026).
- Built Grasshopper auto-generation scripts for recurring structural assemblies, standardizing engineering output across the team and creating a shared information base for successor projects.
- Concurrently supervised four station worksites (Neishi, Jialu, Fangshan, Longxi) on the South-Link Railway Electrification program with Sinotech Engineering Consultants.
- Led a 20-person team delivering Lunping Cultural Landscape Park — Taiwan's first curved glulam structure with a cantilever exceeding 10 m — from design development through construction (2021).
- Collaborated with Kengo Kuma & Associates (KKAA) on the Tsumiki Pavilion, 2018 Taichung World Flora Exposition; coordinated heavy-glulam fabrication programs in North America and imports to Taiwan.

Selected Projects

2023–	Chiayi Railway Elevation — robotic machining of timber structural components , CCI / CECI Engineering Consultants	Chiayi
2024–	Siemens Gamesa offshore-wind horizontal pre-assembly equipment , CCI	Taiwan
2022	NTU Health Center skybridge — envelope digital fabrication , TAKEN / Fei & Cheng Associates	Taipei
2021	Swire Motors service center — envelope packages , TAKEN	Taipei
2020–2021	Lunping Cultural Landscape Park — curved glulam canopy, >10 m cantilever , Fure Jang	Taoyuan
2019–2021	South-Link Railway Electrification — four station sites , Fure Jang / Sinotech	Pingtung & Taitung
2018	Tsumiki Pavilion, Taichung World Flora Exposition , Fure Jang / Kengo Kuma & Associates	Taichung

Education

National Taiwan University of Science and Technology (Taiwan Tech)

Taipei, Taiwan

M.S. IN ARCHITECTURE — DIGITAL FABRICATION

- Thesis: *Digital Fabrication of the Timber Structure in Jiabei Station* — parametric integration of steel–timber hybrid connections and robotic machining workflows (KUKA KR60: milling, slotting, and drilling path tests via the Robots plugin).
- GAS Lab (Prof. Shen-Guan Shih): architectural digital geometry, topology, and robotic fabrication research.

Certifications & Regulatory Approvals

2025	New-construction-material approvals for CLT panels (led team): 1-hour fire-resistance & wind-driven-rain ratings , Ministry of the Interior, Taiwan	Taiwan
2023–	Taiwan distributorship — Stora Enso (mass timber) , Stora Enso	
2024–	Taiwan distributorship — K&S weather-resistant coatings , K&S, Germany	

Skills

Computational Design	Rhino 3D, Grasshopper (parametric BOM & assembly generators), Python (RhinoCommon, AI-assisted), AutoCAD
Robotic Fabrication	KUKA KRL path programming, Robots plugin for Grasshopper, robotic milling / slotting / drilling
Domain	Mass timber (CLT/GLT), steel–timber hybrid connections, LOD 400 BIM coordination, construction PM
Languages	Mandarin Chinese (native), English (professional working proficiency)